

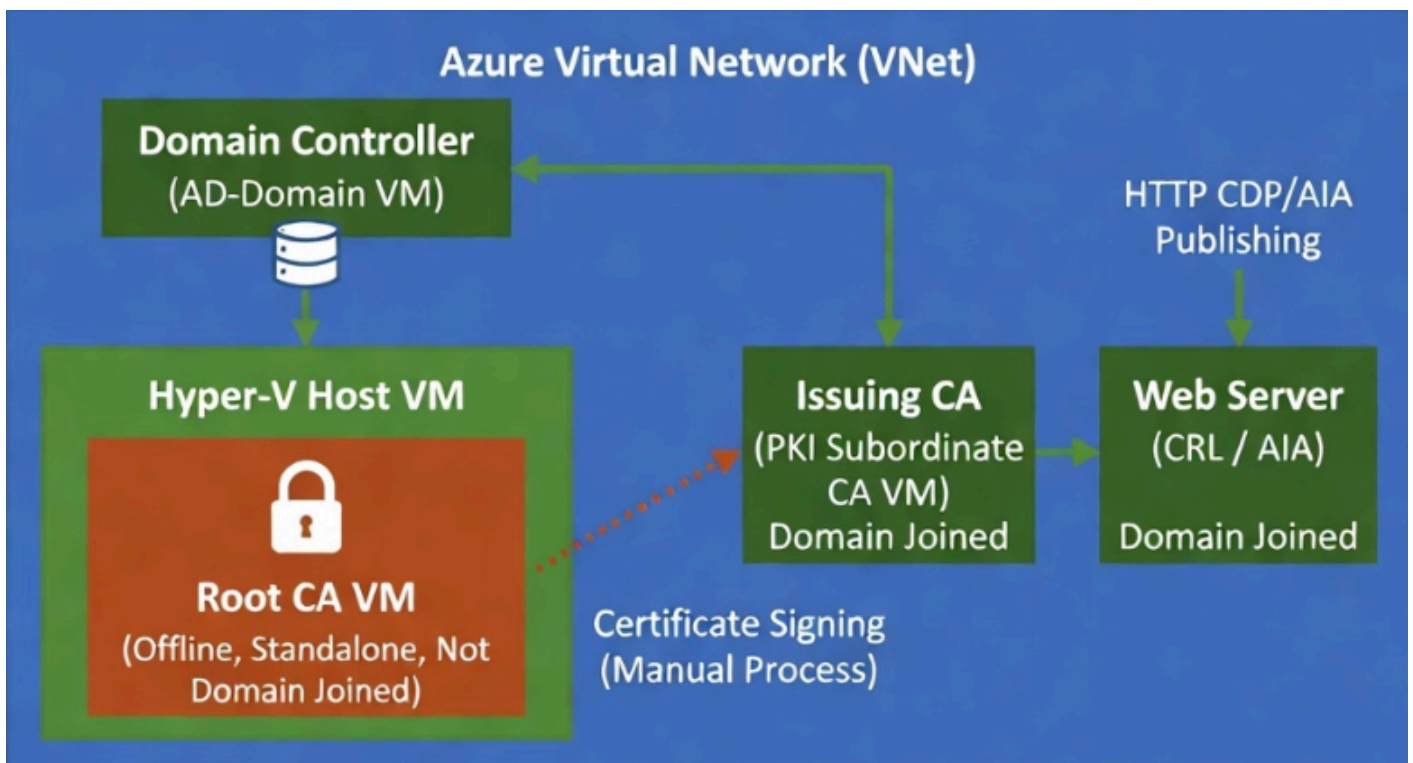
Standard Operating Procedure

Deploying a Two-Tier PKI Hierarchy

Offline Root CA → Enterprise Subordinate (Issuing) CA → IIS CDP/AIA Web Server

Domain: Quantumspace.com

Environment: Azure Free Account / Lab



1. Overview & Purpose

This SOP walks through building a two-tier Public Key Infrastructure (PKI) using Active Directory Certificate Services (AD CS) inside your existing Azure free account lab, on top of the Quantumspace.com Active Directory domain you have already deployed. A two-tier hierarchy separates trust (the Root CA) from day-to-day certificate issuance (the Subordinate/Issuing CA), which is the configuration Microsoft recommends for any environment beyond a single test machine.

Why this step: A single-tier PKI (one CA doing everything) means that if that CA is ever compromised, every certificate it ever issued must be considered compromised. In a two-tier model the Root CA is built once, used briefly to certify the Subordinate CA, and then taken offline (or in Azure's case, deallocated) for the rest of its life — drastically shrinking its attack surface while the Subordinate CA handles routine issuance.

1.1 Target Architecture

Four servers are involved. AD-Domain already exists per your setup. This guide builds the remaining three: Root-CA (Root), Issuing-CA (Issuing), and PKI-WEB-SRV (CDP/AIA web publication point).

VM Name	Role	Domain Join	Network
AD-Domain	Existing AD DS + DNS domain controller for Quantumspace.com (already built)	N/A (is the DC)	Existing VNet/subnet
Root-CA	Standalone Offline Root CA	Workgroup (never domain-joined)	Same VNet, isolated hosted over Hyper-V
Issuing-CA	Enterprise Subordinate (Issuing) CA	Quantumspace.com	Existing VNet/subnet
PKI-WEB-SRV	IIS Web Server – HTTP CDP/AIA publication point	Quantumspace.com	Existing VNet/subnet

Note: Server names, IPs, and the OU/domain suffix used throughout this document are suggestions. Wherever you see Root-CA, Issuing-CA, PKI-WEB-SRV, or AD-Domain, substitute your own existing VM names — what matters is the role each machine plays, not the literal hostname.

1.2 Azure Free Account Considerations

- Azure free-tier credit is limited, so keep VM sizes small (B1s/B2s) and use Azure's auto-shutdown schedule on every VM to avoid burning through credit while a machine sits idle.

- Root-CA (the Root CA) is deliberately never joined to the domain and, in the original on-prem guide, is never even connected to a network. In Azure every VM needs a NIC to be reachable at all, so to isolate the Root-CA have created a VM over Hyper-V which completely isolated from network, and keep the VM turned-off except during the brief windows when you are signing the Subordinate CA's certificate or publishing a new CRL.
- Because there is no physical floppy/USB in Azure, file transfer between the isolated Root-CA and Issuing-CA is done via RDP's built-in "drive/clipboard redirection" (copy-paste or a redirected local drive) instead of removable media — functionally the same air-gap discipline, different mechanism.
- Use a resource group dedicated to this lab so you can pause (deallocate) or tear down the whole PKI environment cleanly.

1.3 Prerequisites

- An existing Active Directory Domain Services forest, Quantumspace.com, with AD-Domain as a domain controller and DNS server.
- An account that is a member of Enterprise Admins (needed to install/configure the Enterprise Subordinate CA and publish objects to AD).
- Three additional Windows Server VMs in the same Azure VNet/subnet as AD-Domain: Root-CA, Issuing-CA, PKI-WEB-SRV (Windows Server 2019 or 2022 recommended).
- All servers patched and time-synchronized against AD-Domain (Kerberos and certificate validity are both time-sensitive).

2. Step 1 – Prepare the IIS Web Server for CDP/AIA Publication

PKI-WEB-SRV will host the public HTTP location where every certificate's CRL Distribution Point (CDP) and Authority Information Access (AIA) fields point. Doing this first means the location already exists and is reachable when both CAs are configured to publish to it.

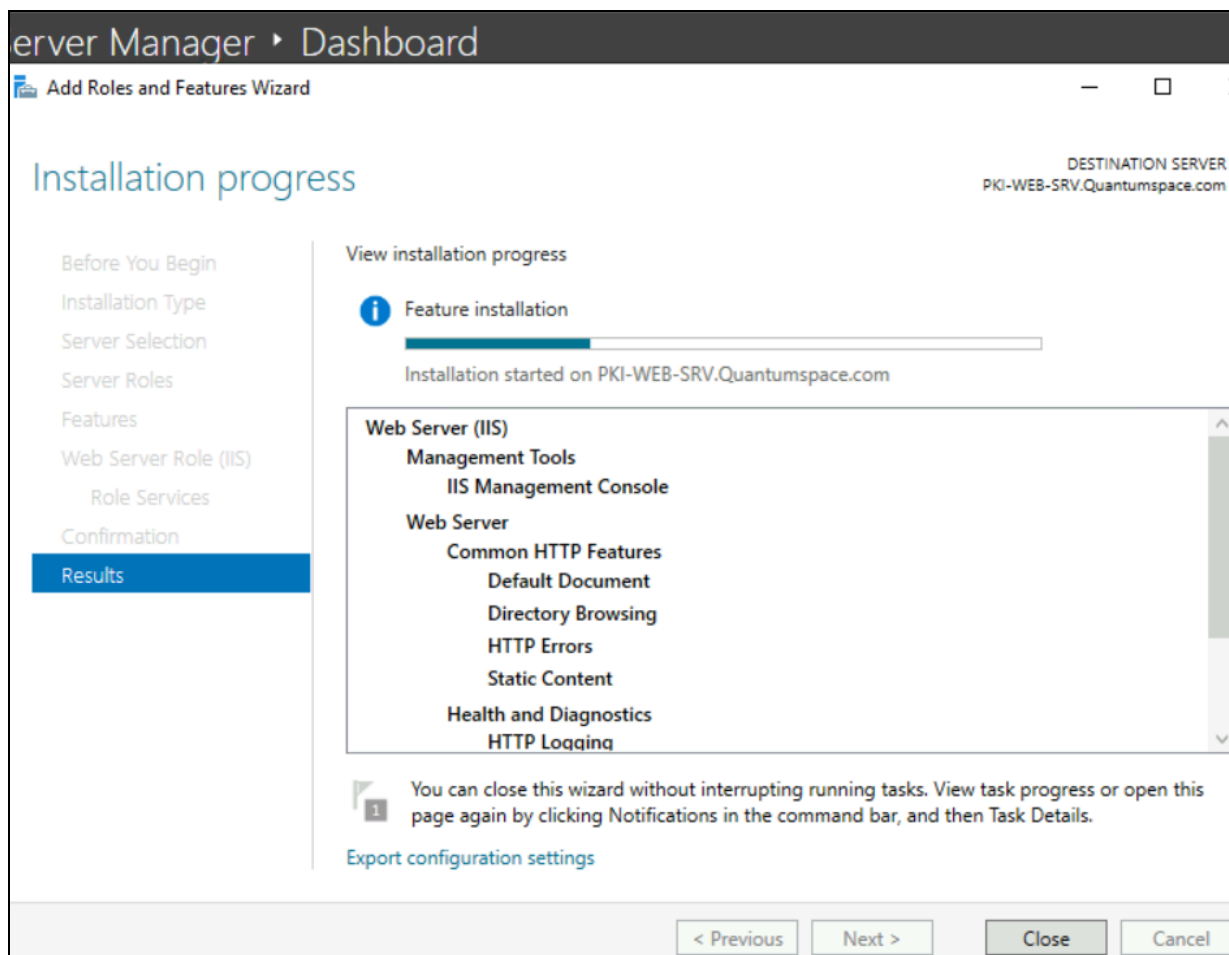
Every certificate embeds a pointer telling the relying party where to fetch (a) the issuing CA's certificate (AIA) and (b) the current revocation list (CDP). Publishing these over plain HTTP — rather than only LDAP — means the certificates remain verifiable by non-domain and external clients, not just machines that can reach your domain controller.

2.1 Join PKI-WEB-SRV to the domain

1. Log on to PKI-WEB-SRV as the local administrator.
2. Open System Properties (sysdm.cpl) → Change → Member of: Domain → type Quantumspace.com → OK.
3. Supply Quantumspace\Administrator credentials (or another Domain Admin account) when prompted.
4. Restart PKI-WEB-SRV when prompted.

2.2 Install the Web Server (IIS) role

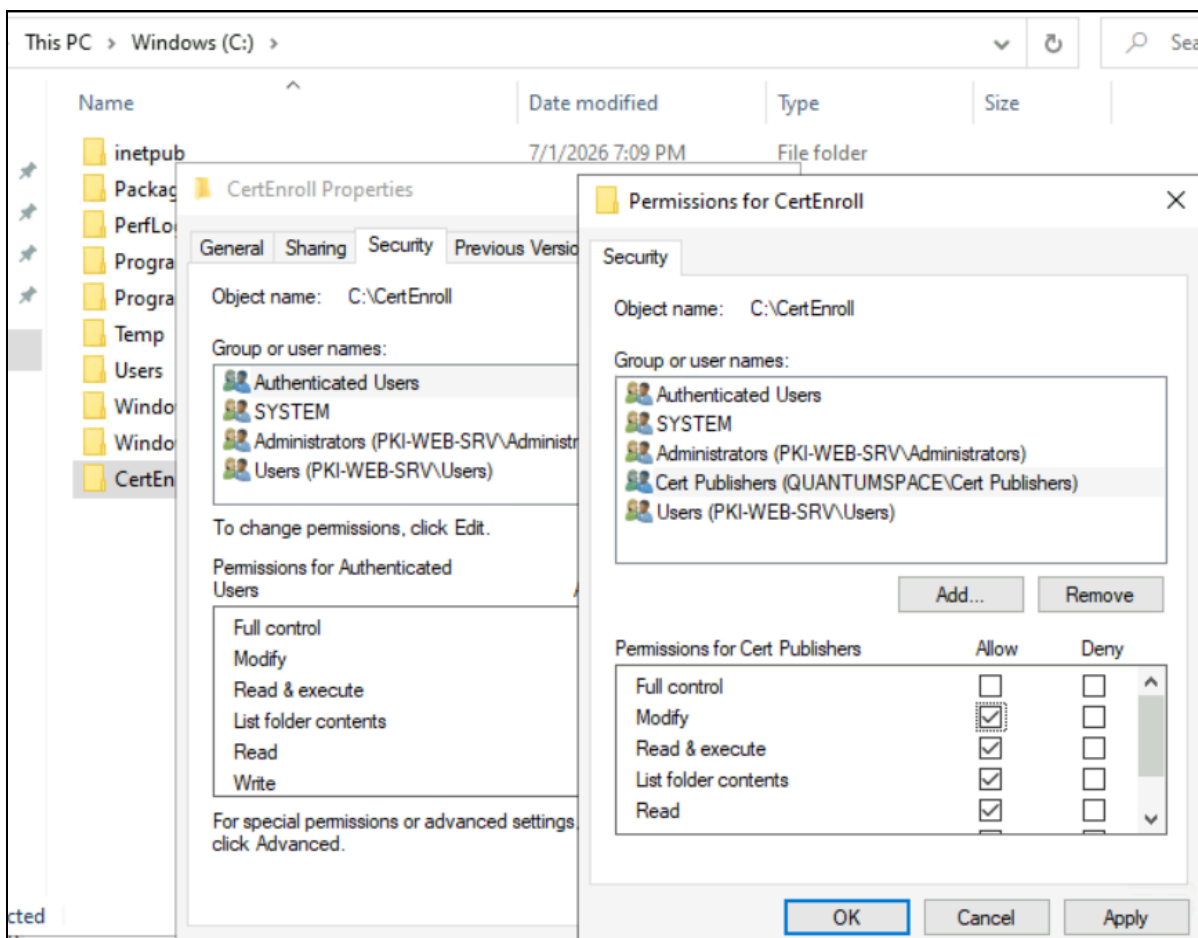
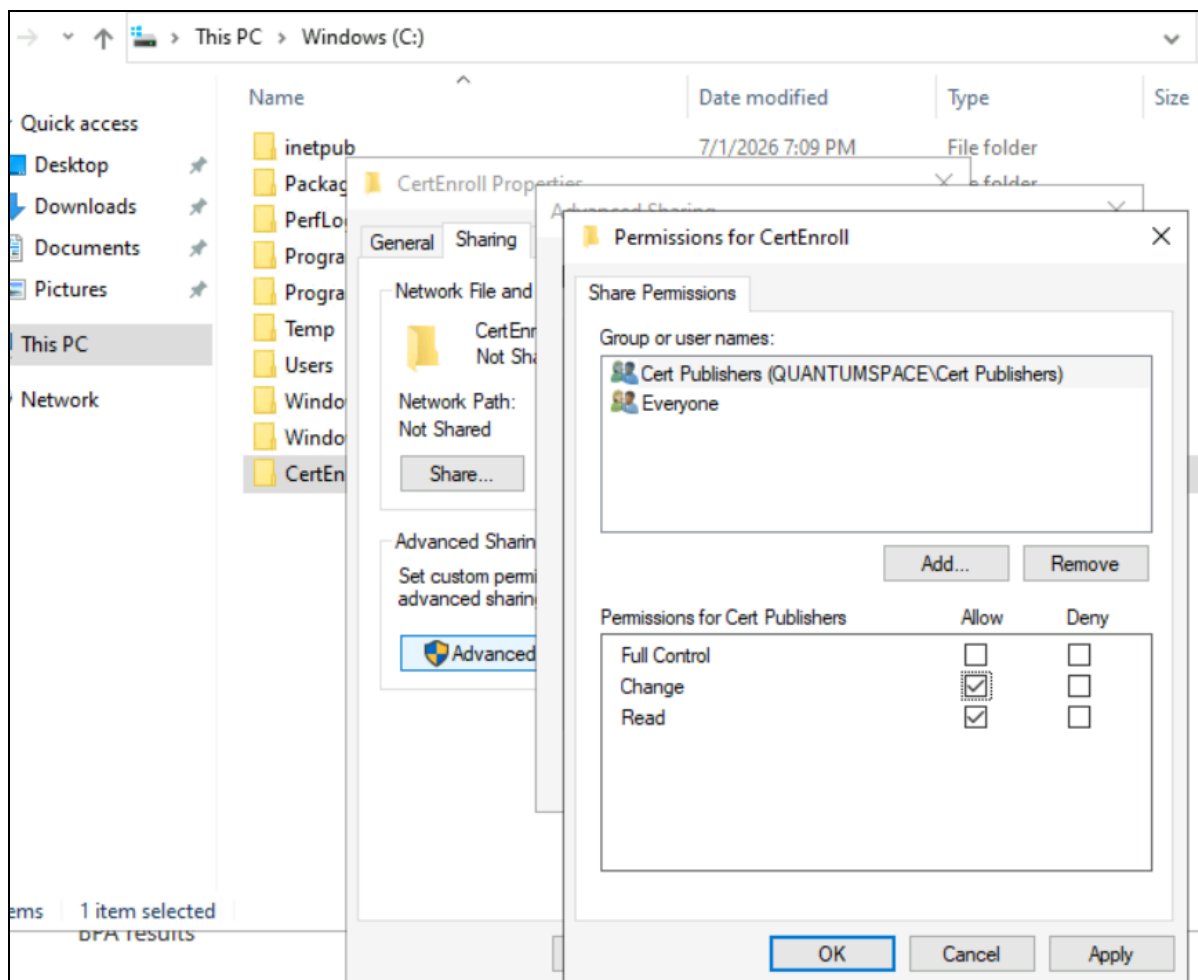
1. Log on to PKI-WEB-SRV.Quantumspace.com as a domain administrator.
2. Open Server Manager → Manage → Add Roles and Features.
3. Server Selection → choose PKI-WEB-SRV → Next.
4. Server Roles → check Web Server (IIS) → Add Features when prompted → Next through Role Services (defaults are fine) → Install.



2.3 Create the CertEnroll folder and grant permissions

The CA servers will remotely copy their certificates and CRLs into this folder, and IIS will serve them out over HTTP.

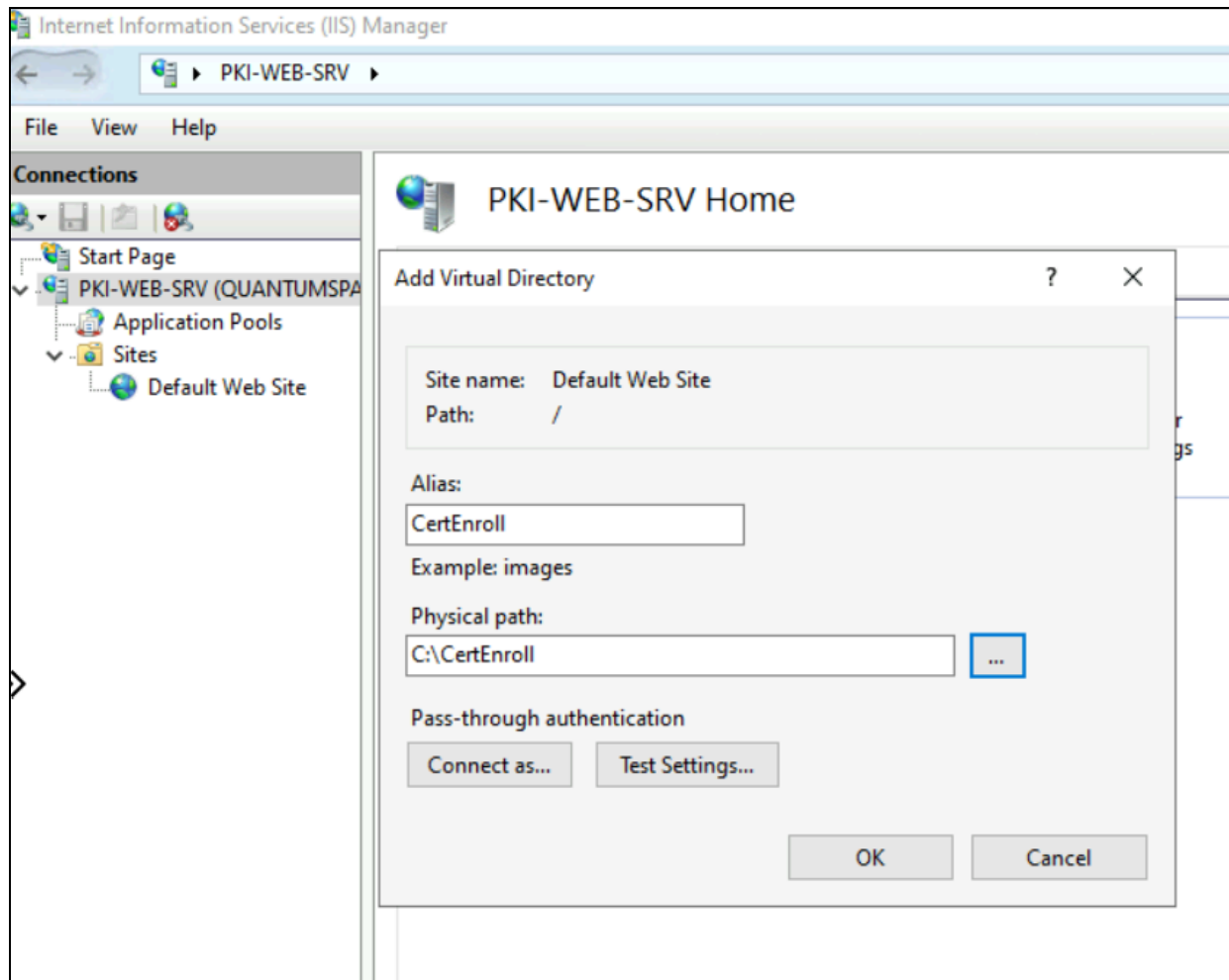
1. On PKI-WEB-SRV, create C:\CertEnroll.
2. Right-click the folder → Properties → Sharing → Advanced Sharing → Share this folder.
3. Permissions → Add → type Quantumspace\Cert Publishers → OK → grant the Cert Publishers group Change (share) permission.
4. Security tab → Edit → Add → Quantumspace\Cert Publishers → grant Modify (NTFS) permission → OK.



Every CA computer account in the domain is automatically a member of Cert Publishers. Granting that group write access, instead of a specific server account means both Root-CA's imported objects and Issuing-CA's automatic publications can land in this folder without per-server ACL maintenance.

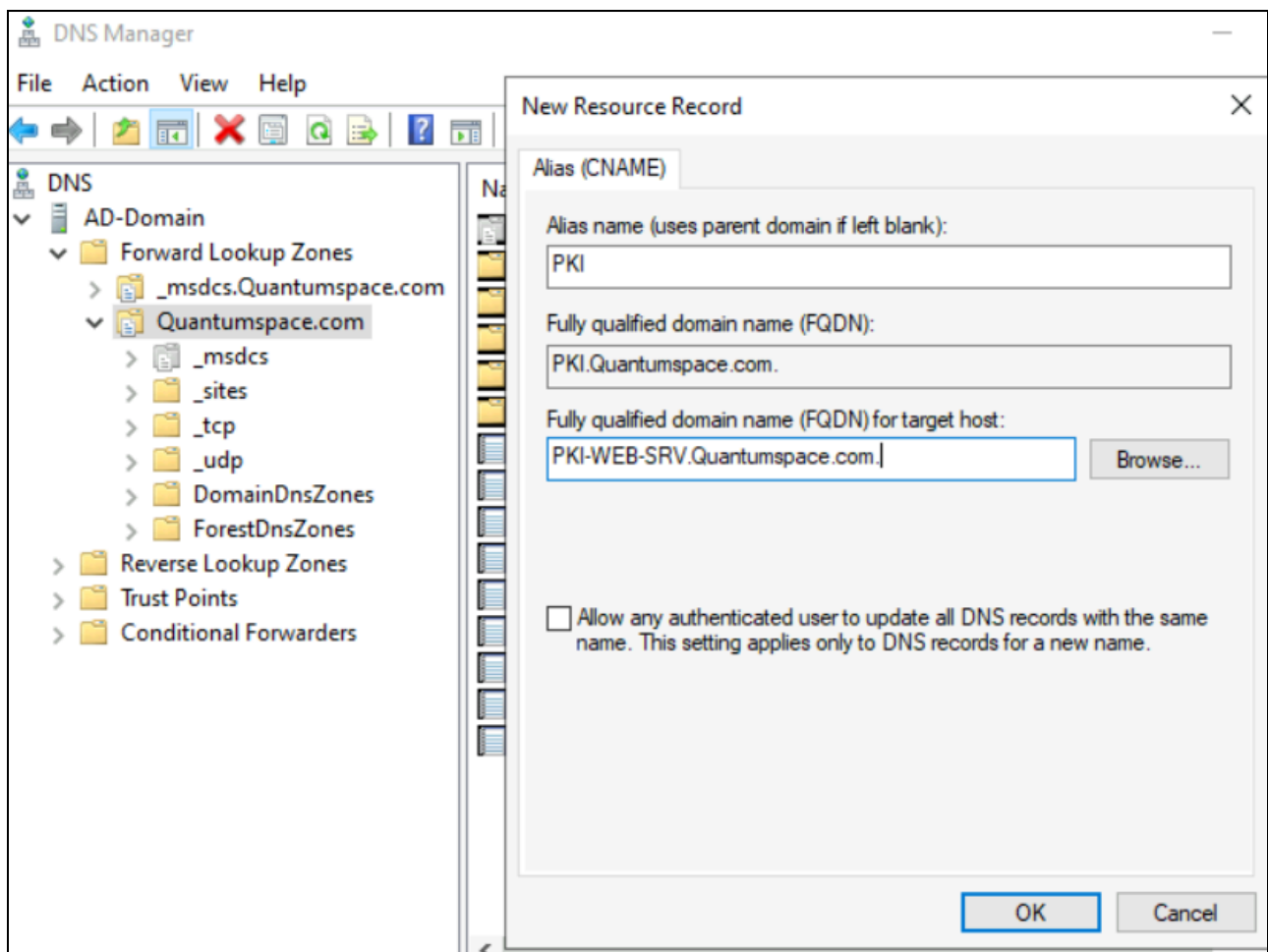
2.4 Create the CertEnroll virtual directory in IIS

1. Open Internet Information Services (IIS) Manager.
2. Expand PKI-WEB-SRV → Sites → right-click Default Web Site → Add Virtual Directory.
3. Alias: CertEnroll. Physical path: C:\CertEnroll → OK.
4. Select the new CertEnroll virtual directory → double-click Directory Browsing → Enable (in the Actions pane).



2.5 Create a DNS alias (CNAME) for the PKI endpoint

1. On AD-Domain, open DNS Manager (dnsmgmt.msc).
2. Expand Forward Lookup Zones → Quantumspace.com → right-click → New Alias (CNAME).
3. Alias name: pki. Fully qualified domain name for target host: PKI-WEB-SRV.Quantumspace.com. (include the trailing dot) → OK.



Using the alias pki.quantumspace.com in every certificate's CDP/AIA field, rather than the literal PKI-WEB-SRV hostname means you can move the web role to a different server, or later put a load balancer in front of several web servers, without reissuing every certificate already in circulation.

When Issuing-CA issues a certificate, it doesn't just say "check for revocation somewhere." It hard-codes an actual URL into the certificate itself, permanently. For example:

CDP: <http://pki.quantumspace.com/CertEnroll/QuantumspaceIssuingCA.crl>

AIA: <http://pki.quantumspace.com/CertEnroll/QuantumspaceIssuingCA.crt>

That URL is baked into every single certificate Issuing-CA hands out, every user cert, every server cert, every device cert. You cannot edit it later. It's part of the signed certificate data.

Why that's a problem if you use the real hostname?

Suppose instead you'd used the literal server name, like this:

CDP: <http://PKI-WEB-SRV.quantumspace.com/CertEnroll/QuantumspaceIssuingCA.crl>

Now imagine two years from now:

- PKI-WEB-SRV dies and you rebuild it as PKI-WEB-SRV2
- Or you outgrow one server and want two web servers behind a load balancer
- Or you just want to rename it for a naming-convention cleanup

Every certificate already issued still points at PKI-WEB-SRV, forever, because that string is frozen inside them. If that hostname stops resolving or stops serving files, every one of those certificates suddenly fails revocation checking — clients try to fetch the CRL, can't reach PKI-WEB-SRV, and treat the certificate as untrustworthy (or refuse it entirely, depending on the application). Your only fix would be to revoke and reissue *every certificate in the environment*. That's a huge, disruptive undertaking.

Why the CNAME alias fixes this?

By baking in pki.quantumspace.com (a DNS alias) instead of the real server name, you've added a layer of indirection. The certificates only ever "know about" the alias — never the actual machine behind it.

So later, when you need to change the underlying infrastructure, you just update *one DNS record*:

pki.quantumspace.com → CNAME → PKI-WEB-SRV2.quantumspace.com

or point it at a load balancer's VIP instead of a single server. Every certificate ever issued — past, present, and future — keeps working without modification, because from their point of view, nothing changed. pki.quantumspace.com still resolves and still serves the CRL/cert files.

3. Step 2 – Deploy the Standalone Offline Root CA (Root-CA)

The Root CA is the ultimate trust anchor for your entire PKI. It is built as a Standalone CA (not Enterprise) and is never domain-joined, because its only job is to sign one thing, the Subordinate CA's certificate and then be

shut down. Keeping it out of Active Directory and off the network the rest of the time is what makes the two-tier model meaningfully more secure than a single CA.

3.1 Prepare the Root-CA VM

- Deploy Root-CA as a workgroup machine (do not join it to Quantumspace.com).

3.2 Create CAPolicy.inf

This file pre-configures renewal and signature settings before AD CS is installed.

1. Log on to Root-CA as the local administrator.
2. Open Notepad and create C:\Windows\CAPolicy.inf with the following content, then save (confirm the .inf extension is kept, not .txt):

[Version]

Signature="\$Windows NT\$"

[Certsrv_Server]

RenewalKeyLength=4096

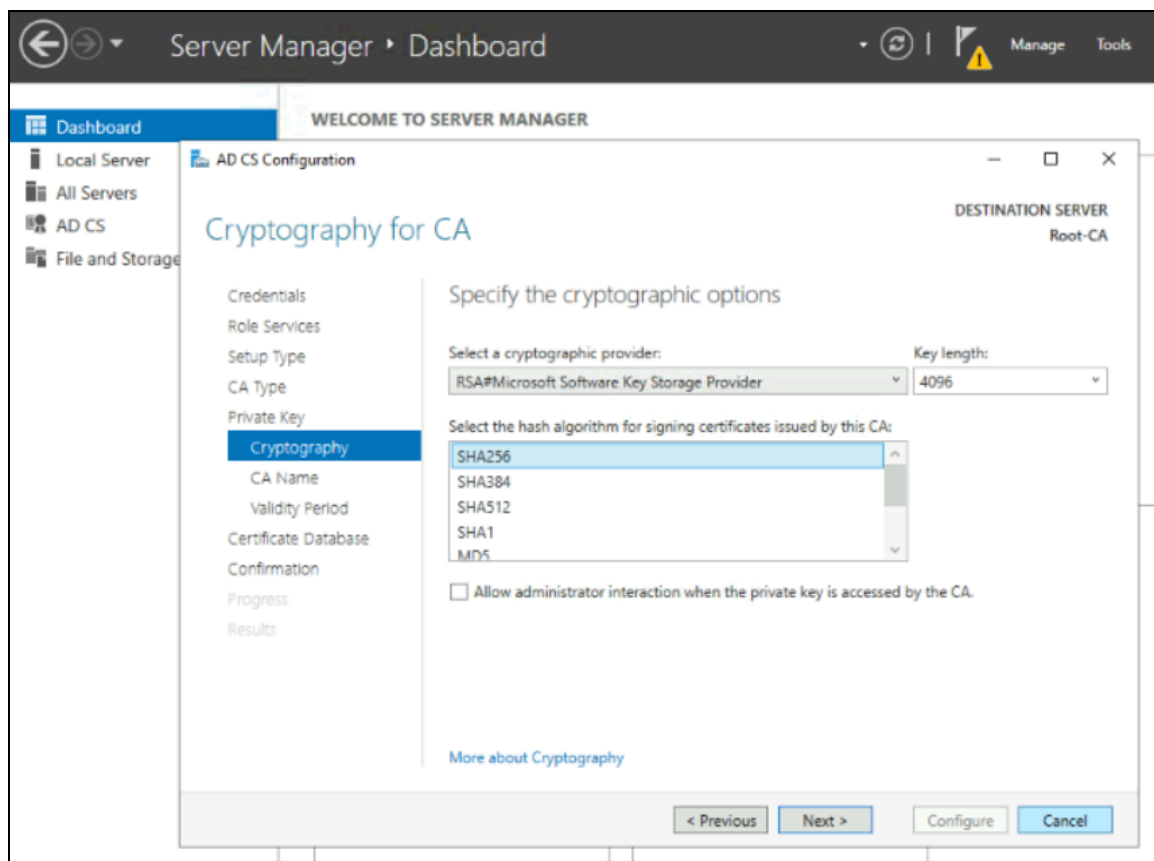
RenewalValidityPeriod=Years

RenewalValidityPeriodUnits=10

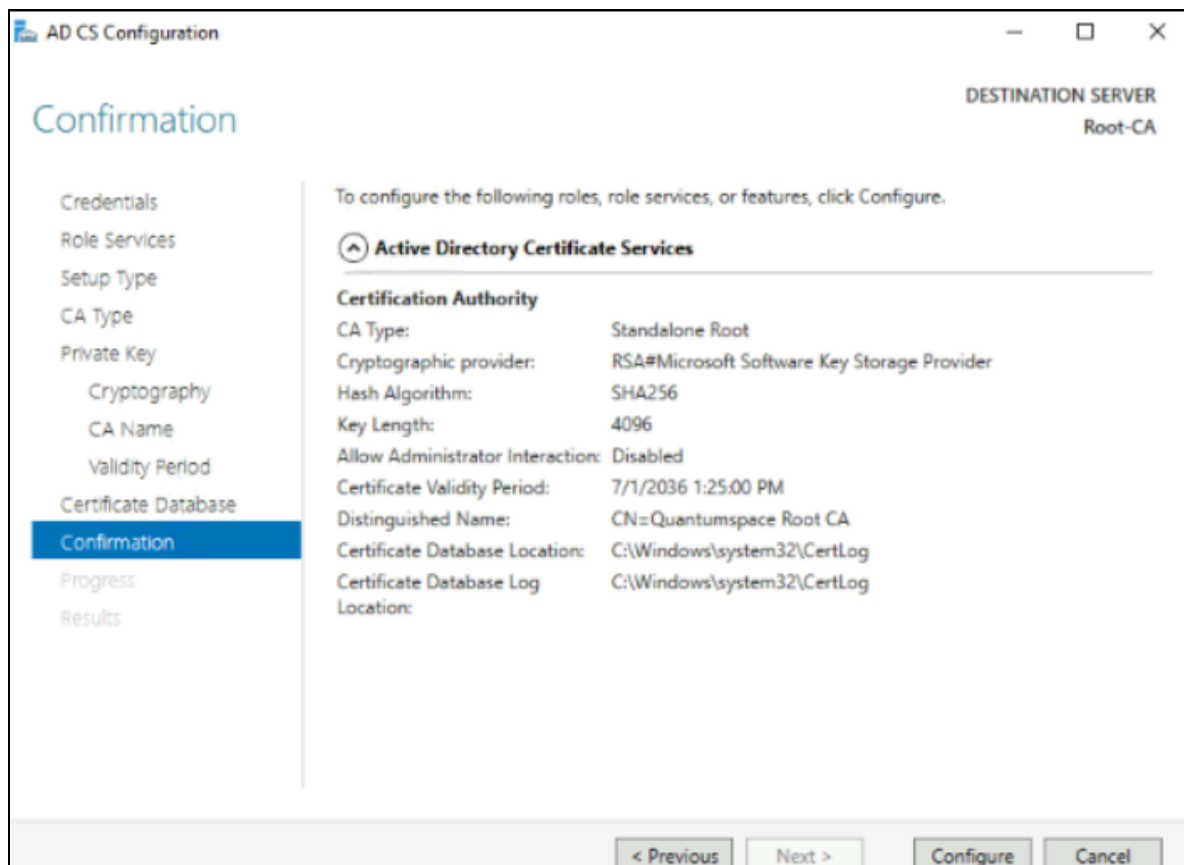
AlternateSignatureAlgorithm=0

3.3 Install and configure the Root CA role

1. Open Server Manager → Add Roles and Features → Server Roles → check Active Directory Certificate Services → Add Features → Next → Role Services: ensure Certification Authority is checked → Install.
2. When installation finishes, click the “Configure Active Directory Certificate Services” link (yellow flag in Server Manager).
3. Credentials page → Next (local Administrator is used since Root-CA is not domain-joined).
4. Role Services → check Certification Authority → Next.
5. Setup Type → Standalone CA → Next (Enterprise is greyed out because Root-CA is not domain-joined — this is expected).
6. CA Type → Root CA → Next.
7. Private Key → Create a new private key → Next.
8. Cryptography → leave defaults (RSA#Microsoft Software Key Storage Provider, SHA256, 4096-bit) for a lab; in production, size the key/hash to your compliance requirements → Next.



1. CA Name → clear the suggested Common Name and enter: Quantumspace Root CA → Next.
2. Validity Period → 10 Years → Next.
3. Certificate Database → leave defaults → Next → Configure → Close.



3.4 Post-installation configuration of the Root CA

Run the following from an elevated Command Prompt on Root-CA. These registry settings define how long CRLs are valid and how long the Root CA's own issued certificates (in practice, just the Subordinate CA cert) remain valid.

```
Certutil -setreg CA\DSConfigDN "CN=Configuration,DC=Quantumspace,DC=com"
```

```
Certutil -setreg CA\CRLPeriodUnits 26
```

```
Certutil -setreg CA\CRLPeriod "Weeks"
```

```
Certutil -setreg CA\CRLDeltaPeriodUnits 0
```

```
Certutil -setreg CA\CRLOverlapPeriodUnits 12
```

```
Certutil -setreg CA\CRLOverlapPeriod "Hours"
```

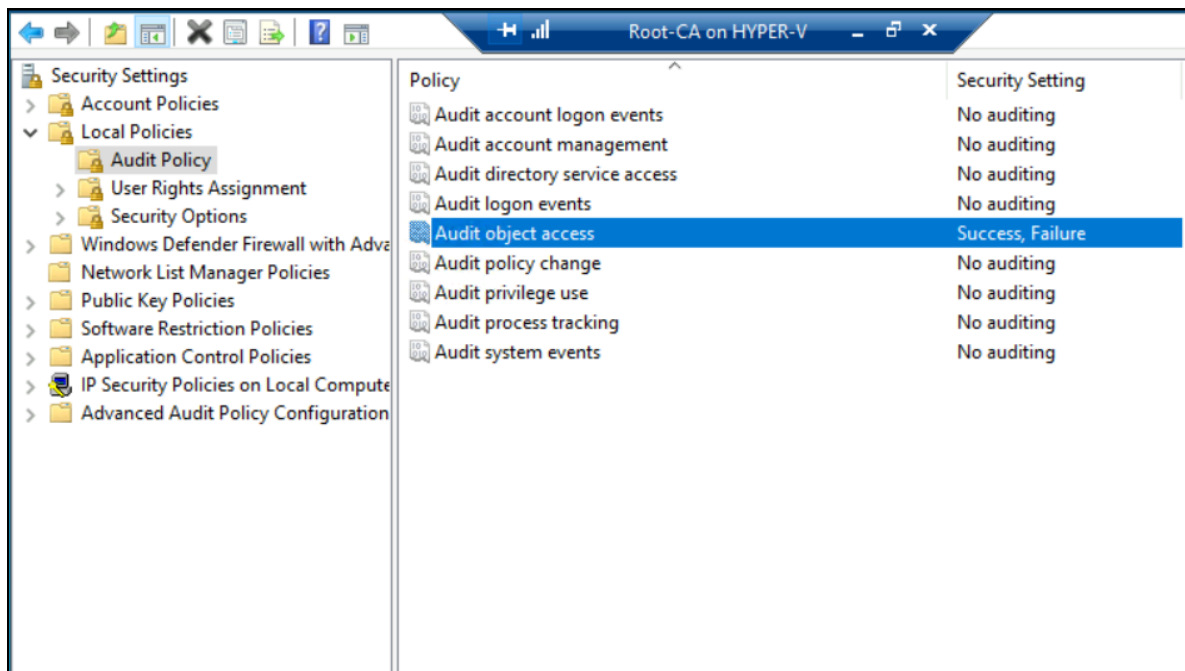
```
Certutil -setreg CA\ValidityPeriodUnits 5
```

```
Certutil -setreg CA\ValidityPeriod "Years"
```

Because the Root CA will be offline almost all the time, its own CRL is set to a long, 6 months validity so it does not expire while the machine is offline. The 5-year Validity Period caps how long a certificate issued by this Root (i.e., the Subordinate CA's certificate) can be valid, it cannot outlive the trust anchor that signed it.

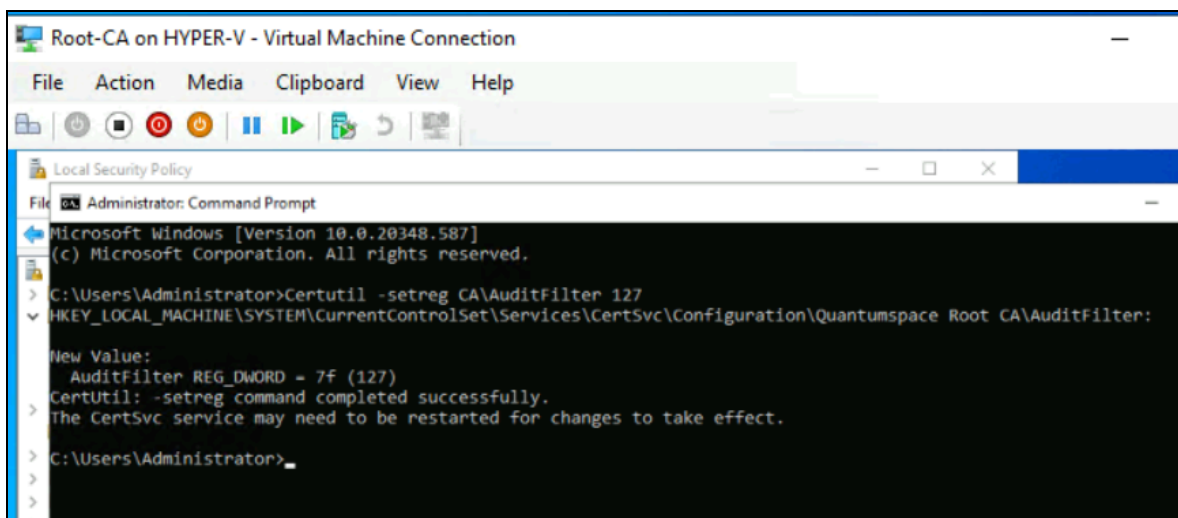
3.5 Enable auditing on the Root CA

1. Open Local Security Policy → Local Policies → Audit Policy → double-click Audit object access → check Success and Failure → OK.



1. From an elevated Command Prompt, run:

```
Certutil -setreg CA\AuditFilter 127
```



A Root CA's private key is the single most sensitive object in the whole hierarchy; auditing every CA-related event gives you a tamper-evident log of who signed what and when, which is critical if you ever need to investigate a compromise.

3.6 Configure AIA and CDP on the Root CA

Run from an elevated Command Prompt on Root-CA:

```
certutil -setreg CA\CACertPublicationURLs
```

```
"1:C:\Windows\system32\CertSrv\CertEnroll\1_%3%4.crt\n2:ldap:///CN=%7,CN=AIA,CN=Public Key  
Services,CN=Services,%6%11\n2:http://PKI.Quantumspace.com/CertEnroll/%1_%3%4.crt"
```

```
certutil -setreg CA\CRLPublicationURLs
```

```
"1:C:\Windows\system32\CertSrv\CertEnroll\3%8%9.crl\n10:ldap:///CN=%7%8,CN=%2,CN=CDP,CN=Public Key  
Services,CN=Services,%6%10\n2:http://PKI.Quantumspace.com/CertEnroll/%3%8%9.crl"
```

```
Restart-Service certsvc
```

```
certutil -crl
```

Because Root-CA is offline, its LDAP publication path will never actually be reachable, it is included for completeness/consistency but the HTTP path via PKI.Quantumspace.com is what every real client will use to find the Root's certificate and CRL. The Root CA's own CDP is intentionally left with no LDAP entry beyond this default; a root's CDP is rarely, if ever, checked in practice.

4. Step 3 – Deploy the Subordinate CA (Issuing-CA)

Issuing-CA is the CA that domain computers and users will actually talk to day-to-day. Because it is Enterprise (domain-integrated), it can auto-enroll certificates via Group Policy and publish directly into Active Directory — capabilities a Standalone CA does not have.

4.1 Join Issuing-CA to the domain

1. Log on to Issuing-CA as the local administrator.
2. System Properties → Change → Member of: Domain → Quantumspace.com → supply Domain Admin credentials → restart.

4.2 Create CAPolicy.inf for Issuing-CA

1. Log on to Issuing-CA.Quantumspace.com as a domain/enterprise administrator.
2. Create C:\Windows\CAPolicy.inf with the following content (replace the sample OID with one your organization owns for anything beyond a lab):

[Version]

Signature="\$Windows NT\$"

[PolicyStatementExtension]

Policies=InternalPolicy

[InternalPolicy]

OID=1.2.3.4.1455.67.89.5

URL=http://PKI.Quantumspace.com/cps.txt

[Certsrv_Server]

RenewalKeyLength=4096

RenewalValidityPeriod=Years

RenewalValidityPeriodUnits=5

LoadDefaultTemplates=0

AlternateSignatureAlgorithm=0

LoadDefaultTemplates=0 stops Windows from auto-publishing the full stock template list the moment the CA comes online. On a real deployment you enable only the specific templates you intend to use, which keeps the enrollable-template list intentional rather than a leftover default.

4.3 Publish the Root CA certificate and CRL

Since Root-CA is isolated, transfer its output to Issuing-CA using RDP drive/clipboard redirection instead of removable media.

1. On Root-CA, locate C:\Windows\System32\CertSrv\CertEnroll\Root-CA_Quantumspace Root CA.crt and Quantumspace Root CA.crl.
2. Copy both files to Issuing-CA via your RDP session's redirected clipboard or local resource mapping.
3. On Issuing-CA, from an elevated Command Prompt, publish them into Active Directory:

```
certutil -f -dspublish "Root-CA_Quantumspace Root CA.crt" RootCA
```

```
certutil -f -dspublish "Quantumspace Root CA.crl" Root-CA
```

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.5256]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\system32>cd C:\Temp

C:\Temp>certutil -f -dsublish "Root-CA_Quantumspace Root CA.crt" RootCA
ldap:///CN=Quantumspace Root CA,CN=Certification Authorities,CN=Public Key Services,CN=Services,CN=Configuration,DC=Quantumspace,DC=com?cACertificate
Certificate added to DS store.

ldap:///CN=Quantumspace Root CA,CN=AIA,CN=Public Key Services,CN=Services,CN=Configuration,DC=Quantumspace,DC=com?cACertificate
Certificate added to DS store.

CertUtil: -dsPublish command completed successfully.

C:\Temp>certutil -f -dsublish "Quantumspace Root CA.crl" Root-CA
ldap:///CN=Quantumspace Root CA,CN=Root-CA,CN=CDP,CN=Public Key Services,CN=Services,CN=Configuration,DC=Quantumspace,DC=com?certificateRevocationList
Base CRL added to DS store.

CertUtil: -dsPublish command completed successfully.

C:\Temp>
```

1. Copy both files into the CDP/AIA share on PKI-WEB-SRV:

```
copy "Root-CA_Quantumspace Root CA.crt" \\PKI-WEB-SRV.Quantumspace.com\C$\CertEnroll\
```

```
copy "Quantumspace Root CA.crl" \\PKI-WEB-SRV.Quantumspace.com\C$\CertEnroll\
```

1. Add the Root certificate/CRL to Issuing-CA's local trust store:

```
certutil -addstore -f root "Root-CA_Quantumspace Root CA.crt"
```

```
certutil -addstore -f root "Quantumspace Root CA.crl"
```

Publishing to Active Directory (dsublish) makes the Root CA trusted domain-wide via Group Policy without touching every workstation by hand. Publishing to the HTTP share makes the same certificate/CRL reachable by devices that cannot query AD/LDAP directly.

TIP: How you can enable Enhanced Session Mode in Hyper-V and use it to copy/paste files between your host and VM

Step 1: Enable Enhanced Session Mode

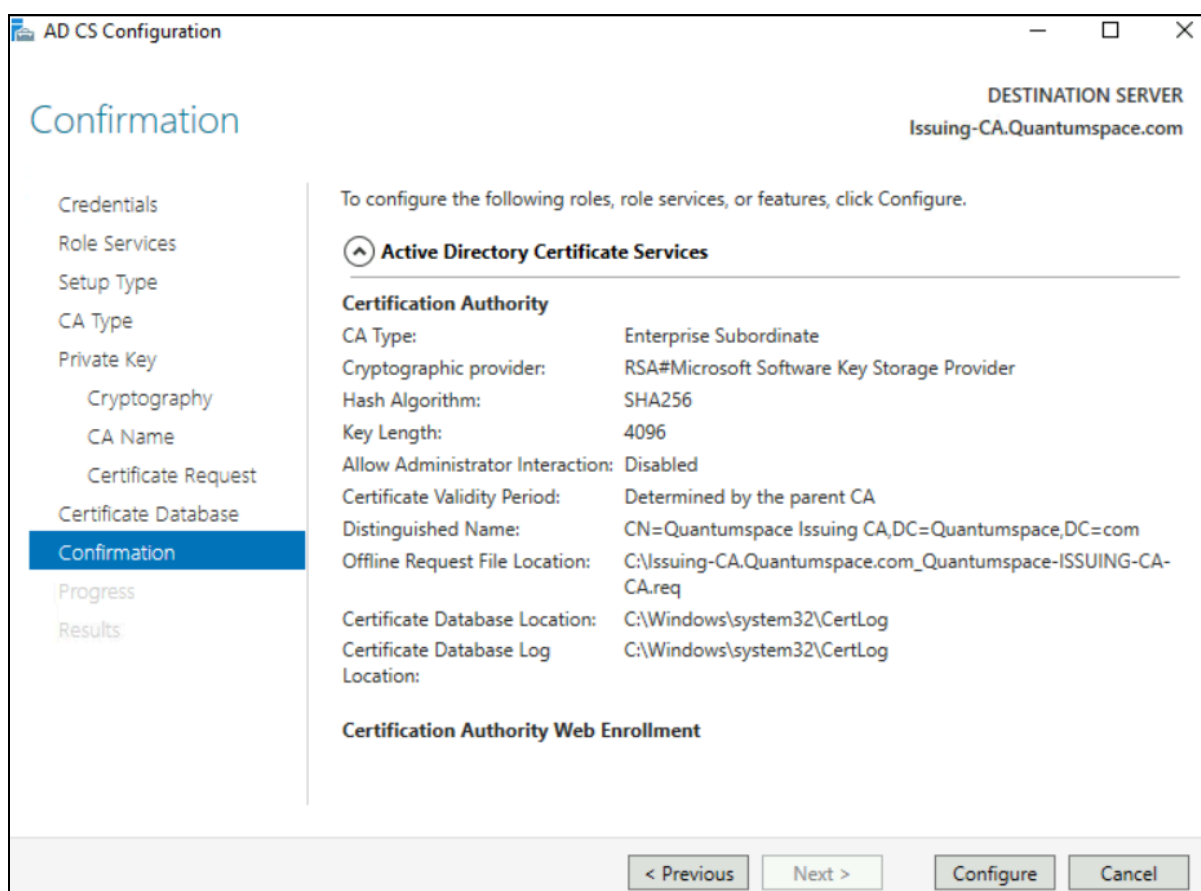
1. Open **Hyper-V Manager** on your host.
2. In the right pane, click **Hyper-V Settings**.
3. Under **Server**, select **Enhanced Session Mode Policy** → check **Allow**.
4. Under **User**, select **Enhanced Session Mode** → check **Use Enhanced Session Mode**.

Step 2: Connect to the VM with Enhanced Session

1. In **Hyper-V Manager**, right-click your VM → **Connect**.
2. When the connection window opens, click the **Show Options** arrow.
3. Go to the **Local Resources** tab.
 - Check **Clipboard** → allows copy/paste text and files.
 - Click **More...** → select host drives (e.g., C:\, D:\) to redirect them into the VM.

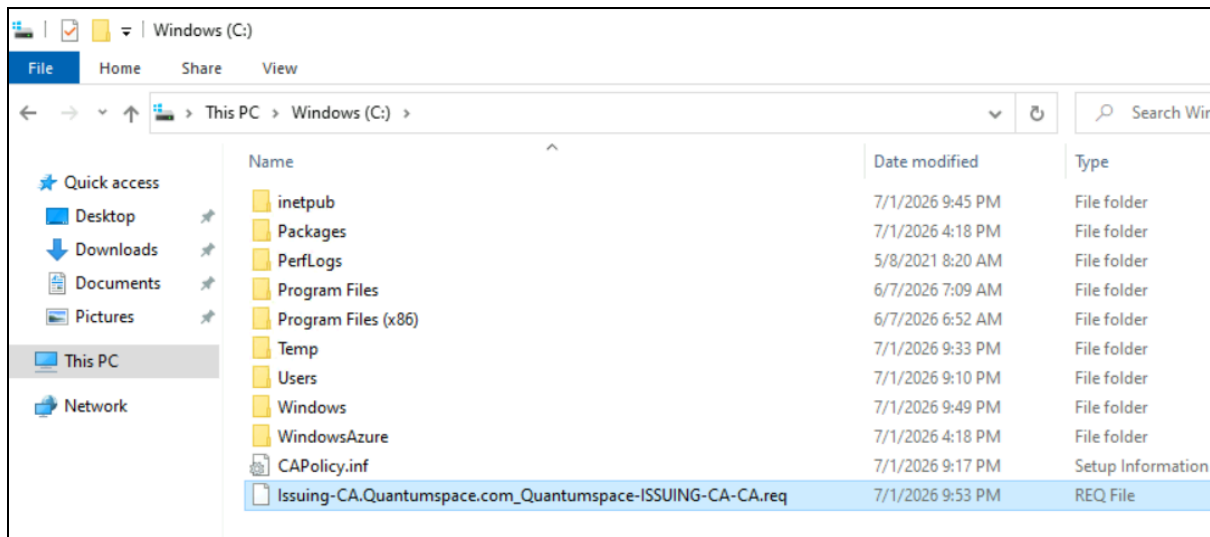
4.4 Install and configure the Subordinate CA role

1. On Issuing-CA, Server Manager → Add Roles and Features → Server Roles → check Active Directory Certificate Services → Add Features → Next.
2. Role Services → check Certification Authority and Certification Authority Web Enrollment (Add Required Role Services when prompted) → Next → Install.
3. Click “Configure Active Directory Certificate Services” in Server Manager.
4. Credentials → confirm an Enterprise Admin account is shown → Next.
5. Role Services → check Certification Authority and Certification Authority Web Enrollment → Next.
6. Setup Type → Enterprise CA → Next (this option requires Enterprise Admin rights and a domain-joined server — both now satisfied).
7. CA Type → Subordinate CA → Next.
8. Private Key → Create a new private key → Next → leave cryptographic defaults → Next.
9. CA Name → clear and enter: Quantumspace Issuing CA (leave the auto-populated Distinguished Name suffix as-is) → Next.
10. Certificate Request → select Save a certificate request to file and manually send it later to a parent CA → Next → leave database defaults → Configure → Close.



Note: You will see an expected warning that the AD CS installation is incomplete until the certificate request is signed by the Root CA. That is normal at this stage.

1. Locate the generated request, typically C:\Issuing-CA.Quantumspace.com_Quantumspace Issuing CA.req, and copy it to Root-CA via RDP redirection.



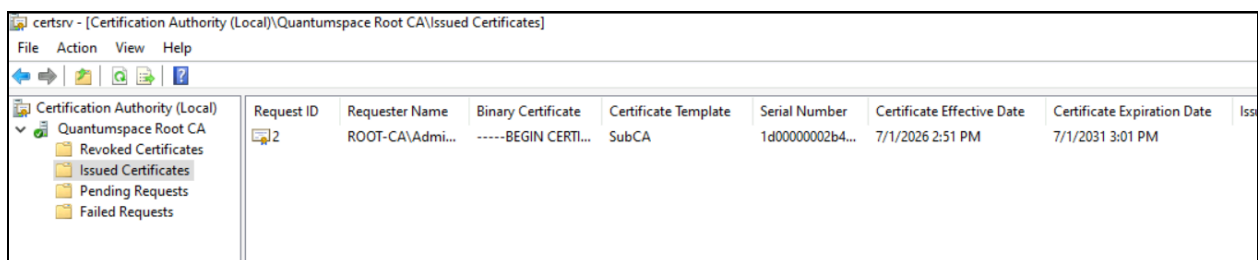
5. Step 4 – Issue the Subordinate CA Certificate on the Root CA

This is the one moment the two CAs need to interact: the Root CA reviews and signs the request that makes Issuing-CA a legitimate, trusted issuing authority under it. This is also the only reason Root-CA needs to be powered on after its initial build.

1. Power on Root-CA and log on as Root-CA\Administrator.
2. Copy the .req file from Issuing-CA onto Root-CA via RDP redirection.
3. From an elevated Command Prompt, submit the request (note the RequestID that is returned):

```
certreq -submit "Issuing-CA.Quantumspace.com_Quantumspace Issuing CA.req"
```

1. If prompted with a Certification Authority List dialog, select Quantumspace Root CA → OK.
2. Open the Certification Authority console (certsrv.msc) → Quantumspace Root CA → Pending Requests → right-click the request → All Tasks → Issue.
3. Back at the Command Prompt, retrieve the now-issued certificate (replace 2 with your actual RequestID):



```
certreq -retrieve 2 "Issuing-CA.Quantumspace.com_Quantumspace Issuing CA.crt"
```

1. Copy the resulting .crt file back to Issuing-CA via RDP redirection.
2. Power off / deallocate Root-CA again once the certificate has been retrieved.

This is the security payoff of the whole design: as soon as this signature is issued, the Root CA has nothing further to do until the Subordinate CA's certificate is due for renewal (in this design, up to 5 years away). Every day it sits deallocated is a day it cannot be attacked over the network.

6. Step 5 – Install the Subordinate CA Certificate and Complete Configuration

6.1 Install the certificate and start the service

1. On Issuing-CA, open the Certification Authority console.

2. Right-click Quantumspace Issuing CA → All Tasks → Install CA Certificate.
3. Browse to the .crt file copied from Root-CA (ensure the file-type filter shows All Files) → Open.
4. Right-click Quantumspace Issuing CA → All Tasks → Start Service.

6.2 Configure CRL, delta CRL, and validity periods

`Certutil -setreg CA\CRLPeriodUnits 1`

`Certutil -setreg CA\CRLPeriod "Weeks"`

`Certutil -setreg CA\CRLDeltaPeriodUnits 1`

`Certutil -setreg CA\CRLDeltaPeriod "Days"`

`Certutil -setreg CA\CRLOverlapPeriodUnits 12`

`Certutil -setreg CA\CRLOverlapPeriod "Hours"`

`Certutil -setreg CA\ValidityPeriodUnits 1`

`Certutil -setreg CA\ValidityPeriod "Years"`

Unlike the Root, this CA is online continuously, so its CRL is refreshed weekly with a daily delta CRL — revoked certificates propagate to relying parties quickly. The 1-year issued-certificate validity is kept of the Issuing CA's own 5-year certificate lifetime, a standard PKI practice so no certificate can ever outlive the CA that issued it.

A CRL is the full list of every certificate this CA has ever revoked — published weekly here, so it's the "big stable snapshot."

A Delta CRL is a small list of only the *new* revocations since that last full CRL — published daily, so revocations (like a stolen laptop or fired employee) get seen by clients within a day instead of waiting up to a week for the next full CRL.

You need both because publishing the entire full list every day is wasteful (it just keeps growing), but checking revocation only once a week is too slow for anything urgent — the delta CRL gives you fast updates cheaply, while the full CRL stays the authoritative complete record.

The overlap period (12 hours) just makes the new CRL go live a bit before the old one expires, so there's never a gap where a client can't find *any* valid CRL. Validity Period is unrelated — it just sets how long certificates issued by this CA (to users/computers) last before they need renewal.

6.3 Enable auditing

`Certutil -setreg CA\AuditFilter 127`

Note: Because Issuing-CA is domain-joined, you may alternatively enable Audit object access through a Group Policy Object linked to the CA's OU instead of Local Security Policy.

6.4 Configure AIA and CDP for Issuing-CA

```
certutil -setreg CA\CACertPublicationURLs
```

```
"1:C:\Windows\system32\CertSrv\CertEnroll\1_%3%4.crt\n2:ldap:///CN=%7,CN=AIA,CN=Public Key  
Services,CN=Services,%6%11\n2:http://PKI.Quantumspace.com/CertEnroll/%1_%3%4.crt"
```

```
certutil -setreg CA\CRLPublicationURLs
```

```
"65:C:\Windows\system32\CertSrv\CertEnroll\3%8%9.crl\n79:ldap:///CN=%7%8,CN=%2,CN=CDP,CN=Public Key  
Services,CN=Services,%6%10\n6:http://PKI.Quantumspace.com/CertEnroll/%3%8%9.crl\n65:\\PKI-WEB-  
SRV.Quantumspace.com\CertEnroll\3%8%9.crl"
```

```
net stop certsvc && net start certsvc
```

```
certutil -crl
```

1. Copy the Issuing CA's own certificate out to the HTTP share so it is retrievable from the AIA URL:

```
copy "C:\Windows\System32\certsrv\certenroll\Issuing-CA.Quantumspace.com_Quantumspace Issuing CA.crt" \\PKI-WEB-  
SRV.Quantumspace.com\C$\CertEnroll\
```

The extra file-system CDP entry (\\PKI-WEB-SRV...\CertEnroll) is what lets Issuing-CA auto-publish every new CRL straight to the web server over the network each time it is regenerated, using the Cert Publishers permissions granted back in Step 1 — no manual copy step is needed going forward.

Whenever you generate a new CRL (which happens automatically, weekly full CRL + daily delta CRL from the previous step), also copy it directly over the network to this UNC path on the web server.

So unlike the certificate (a one-time file), the CRL changes constantly, every day (delta) and every week (full). You obviously don't want to manually copy a new CRL file to the web server every single day by hand. So the CA does it *for you*, automatically, every time it regenerates a CRL, using that network path you configured.

Why does this work without you doing anything? Because back in Step 1, you gave the Cert Publishers AD group write/modify permission on that C:\CertEnroll share and folder. The Issuing CA's computer account is automatically a member of Cert Publishers. So when the CA tries to write a new CRL file into \\PKI-WEB-SRV...\CertEnroll, it already has permission to do so, no prompts, no manual step, it just works going forward.

7. Step 6 – Verify PKI Hierarchy Health

7.1 Enterprise PKI (PKIView) check

1. On Issuing-CA, run pkiview.msc, or in Server Manager open Active Directory Certificate Services → Enterprise PKI.
2. Right-click Enterprise PKI → Manage AD Containers, and confirm on each tab (NTAuthCertificates, AIA Container, CDP Container, Certification Authorities Container, Enrollment Services Container) that both CA certificates and CRLs show a status of OK.

PKIView cross-checks every URL embedded in your certificates against what is actually reachable, catching typos in pki.quantumspace.com paths or missing share permissions before an end user ever hits the problem.

Delta CRL "Unable to Download" in PKIView

If only the DeltaCRL HTTP location shows *Unable to Download* (filename containing a +), while everything else is OK:

1. **Confirm the file exists** on PKI-WEB-SRV at C:\CertEnroll\<CA Name>+.crl. If missing, run certutil -crl on the Issuing CA to force republish.
2. **Check double-escaping is enabled** on PKI-WEB-SRV:

appcmd list config "Default Web Site" /section:requestFiltering

Look for allowDoubleEscaping="false" or missing.

3. **Re-apply the fix:**

appcmd set config "Default Web Site" /section:system.webServer/security/requestFiltering - allowDoubleEscaping:True

iisreset

1. **Test directly in browser:**

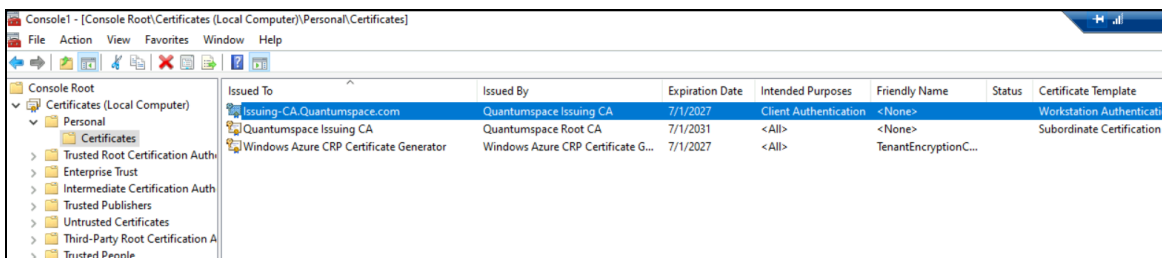
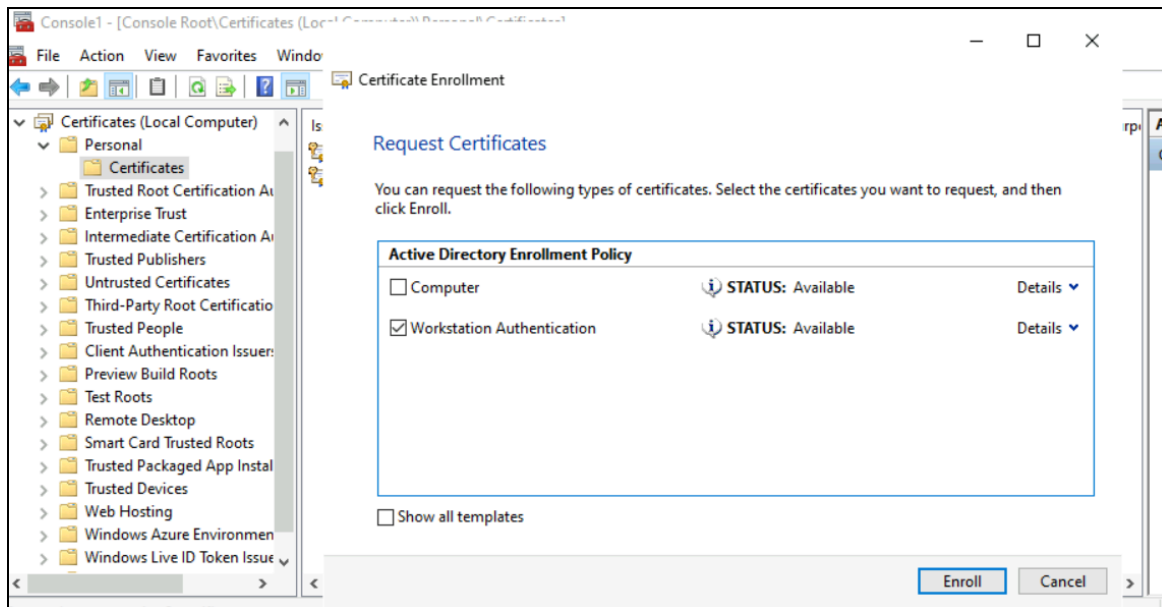
http://pki.quantumspace.com/CertEnroll/<CA Name>+.crl

- 404.11 URL Double Escaped → double-escaping setting didn't persist; redo step 3.
- Plain 404 → file missing; redo step 1.
- Downloads fine → refresh PKIView (right-click **Enterprise PKI** → **Refresh**).

Root cause: The + in delta CRL filenames triggers IIS's default double-escaping security filter, which blocks the request unless explicitly allowed. This setting can fail to persist after an IIS reset/reconfig, so it's worth re-verifying if the issue resurfaces.

7.2 Issue a test certificate and validate the chain

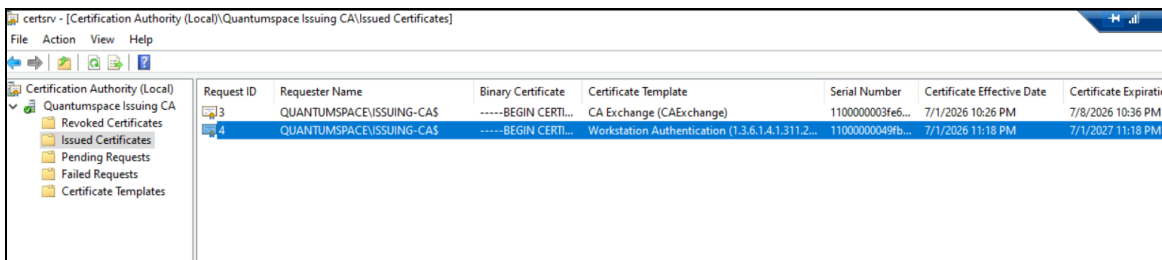
1. On Issuing-CA, in the Certification Authority console, right-click Certificate Templates → New → Certificate Template to Issue → select Workstation Authentication → OK.
2. On a domain-joined client (or Issuing-CA itself for a quick test), open mmc → Add/Remove Snap-in → Certificates (Computer account) → expand Personal → right-click → All Tasks → Request New Certificate → enroll for Workstation Authentication.

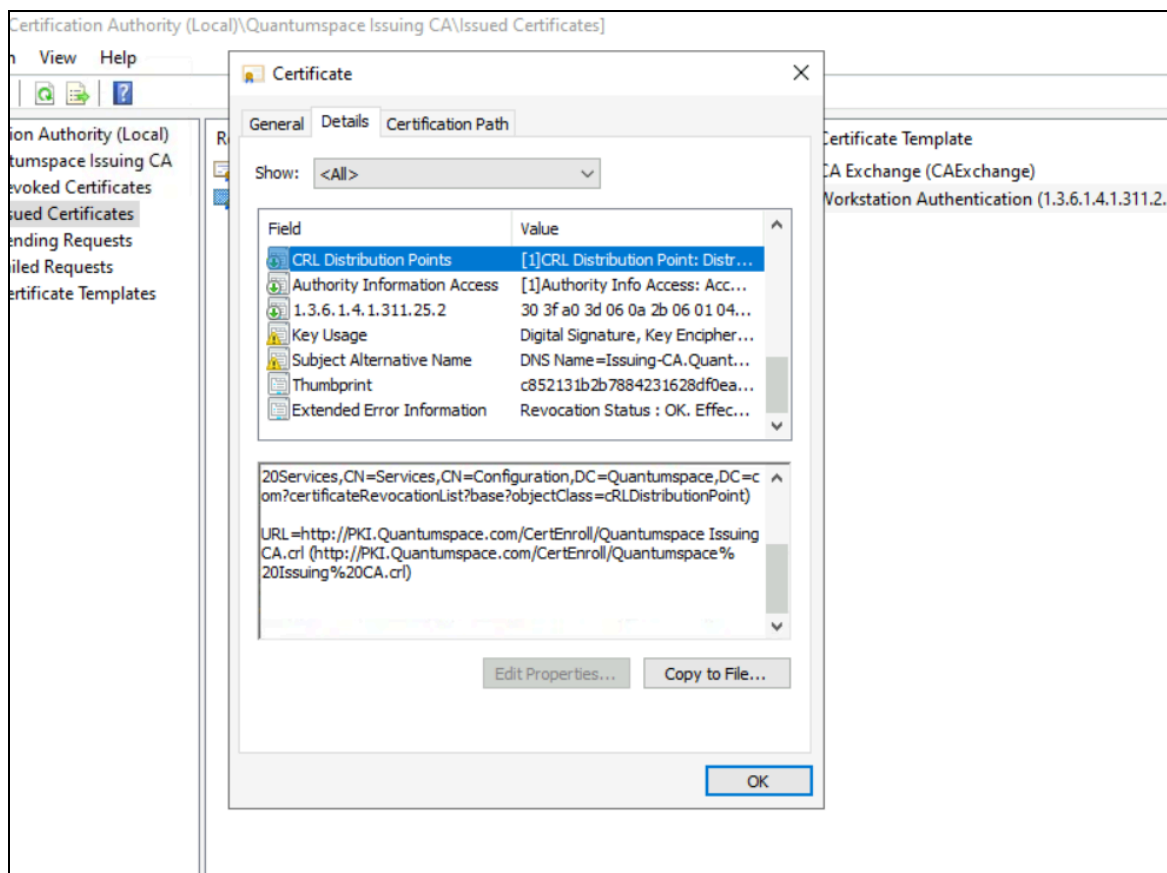


1. Export the resulting certificate (no private key) to a .cer file, then from a Command Prompt run:

```
certutil -verify -urlfetch C:\test.cer
```

1. Confirm the output shows the full chain resolving to Quantumspace Root CA with revocation checking succeeding via the HTTP CDP.



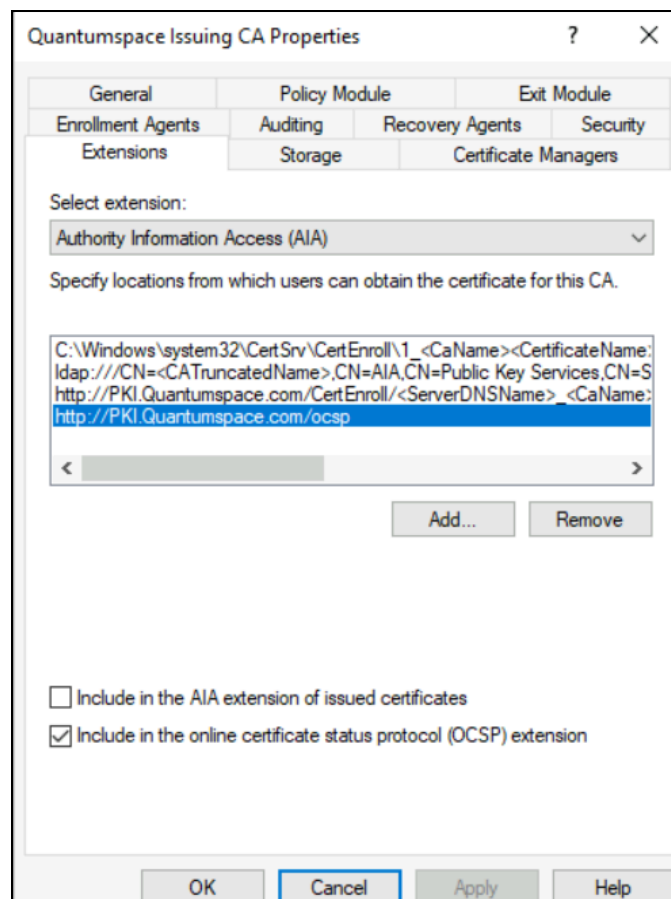


This end-to-end test proves every earlier step actually works together: a client with no direct line to Root-CA or Issuing-CA can still build trust and check revocation purely from what you published to PKI.Quantumspace.com.

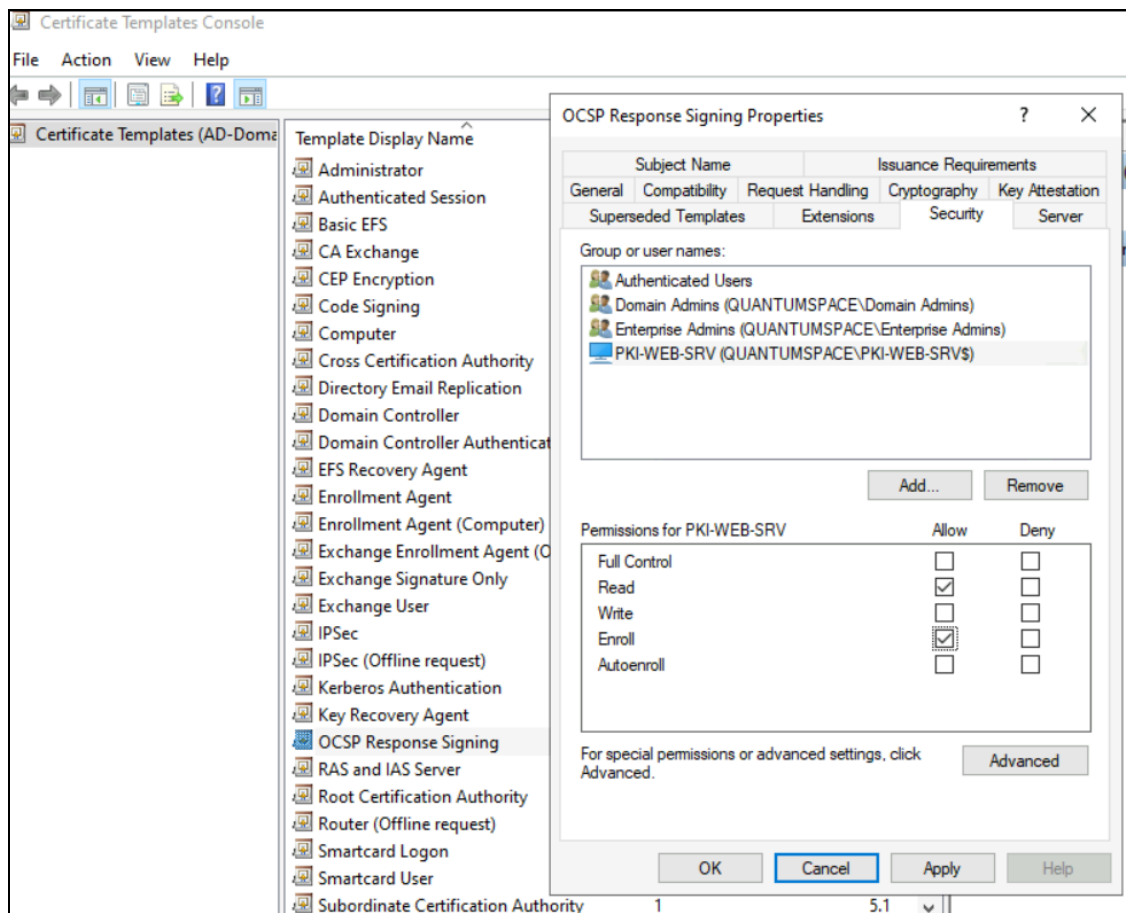
8. Optional – Online Responder (OCSP)

An Online Responder answers real-time “is this one certificate still valid?” queries instead of forcing clients to download the entire CRL.

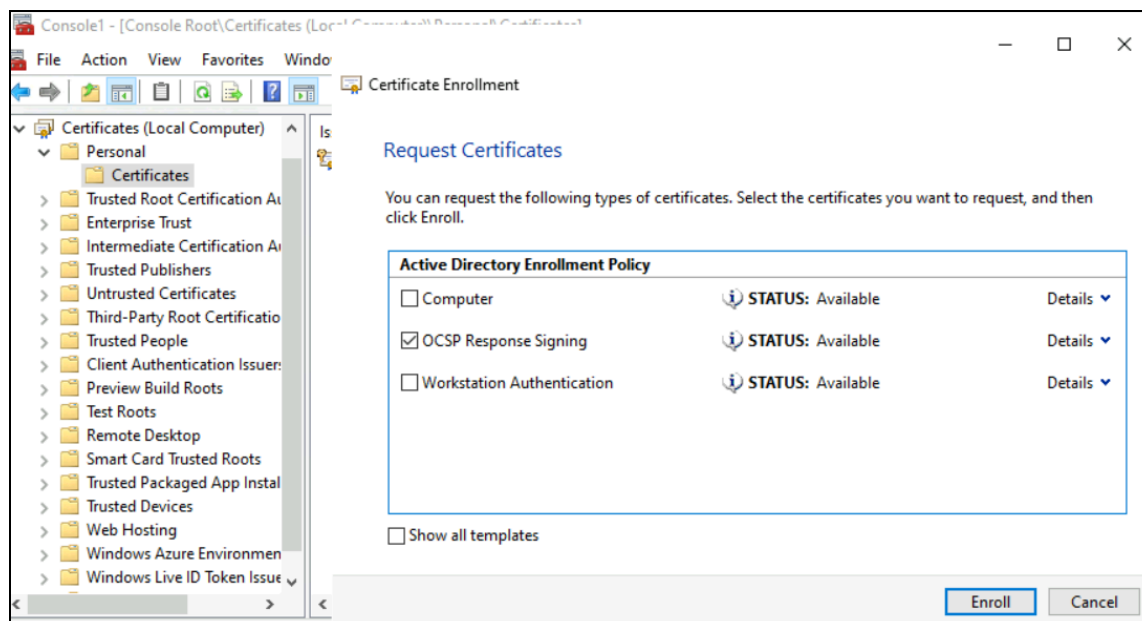
- Install the Online Responder role service on PKI-WEB-SRV (Server Roles → Active Directory Certificate Services → Online Responder).
- On Issuing-CA, add <http://PKI.Quantumspace.com/ocsp> to the Issuing CA's AIA extension and select “Include in the online certificate status protocol (OCSP) extension” only.



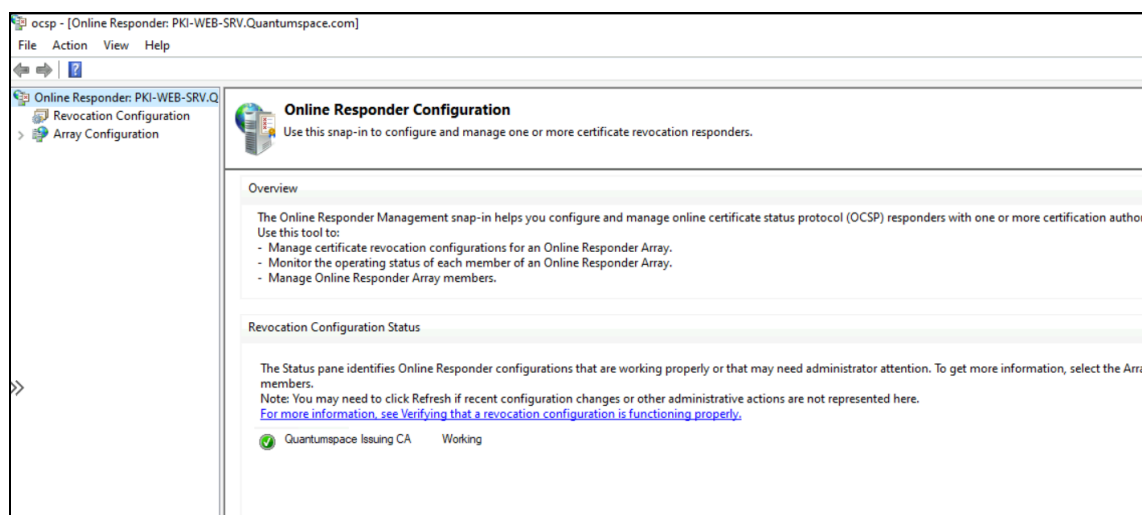
- Give permission to PKI-WEB-SRV to enroll with OCSP template.



- Issue the OCSP Response Signing template to PKI-WEB-SRV's computer account and enroll it.



- In the Online Responder console on PKI-WEB-SRV, add a Revocation Configuration pointing at Quantumspace Issuing CA.



9. Reference Links

Official Microsoft documentation referenced throughout this SOP:

- [AD CS: Two-Tier PKI Hierarchy Deployment \(source guide\)](#).
- [Certutil](#)
- [Install-AdcsCertificationAuthority \(PowerShell\)](#).
- [Install-AdcsWebEnrollment \(PowerShell\)](#).
- [IIS: Create a Virtual Directory](#).